

Sri Lanka Institute of Information Technology

Data Warehousing and Business Intelligence IT3021

Assignment 2

2026

IT23274648 – Wijesuriya W.A.A.I

Year 3 Semester 2

Contents

Introduction 4
Tools & Technologies 4
Data Warehouse Overview: 4
Fact Table: 4
Dimension Tables: 4

Conceptual ER Diagram 6
SSAS Cube Implementation 7
Creating the SSAS Project: 7
Creating the Data Source (DS_EDW): 8
Creating the Data Source View (DSV_EDW): 8
Creating the Cube (Cube_EDW): 9
Adding Attributes to Dimensions: 11
DimUser: 11
DimProduct: 12
DimDistributionCenter: 12
DimDate: 13
Creating Hierarchies: 13
Deploying the Cube: 14
Demonstration of OLAP operations 16
1.Roll-up / Drill-down: 16
Process: 16
2. Slice: 17
Process: 17
3. Dice: 18
Process: 18
18
4. Pivot: 18
Process: 19
Power BI Reports: 19
Creating Required Reports 20
Report 1: Matrix visual 20

Report 2: Cascading Slicers with Multiple Visuals: 21
Process: 21
Report 3: Drill-Down Report: 21
Process: 21
Report 4: Drill-through Report: 23

Introduction

This assignment focuses on creating a Business Intelligence (BI) solution with a data warehouse built in Assignment 01. The goal is to create a multidimensional model (cube), perform OLAP operations, and produce analytical reports using tools like SSAS, Excel, and Power BI.

The dataset used is the E-commerce dataset, which includes transactional data such as orders, users, products, and distribution centers.

Tools & Technologies Used: -

- **SQL Server Management Studio (SSMS)** – Database management
- **SQL Server Integration Services (SSIS)** – ETL process
- **SQL Server Data Tools (SSDT)** – Cube creation (SSAS)
- **SQL Server Analysis Services (SSAS)** – Multidimensional model
- **Microsoft Excel** – PivotTable analysis
- **Power BI Desktop** – Interactive dashboards

Data Warehouse Overview:

The EcommerceDW follows a star schema architecture, consisting of one central fact table surrounded by multiple dimension tables.

Fact Table:

- **FactOrders –**

  Measures:

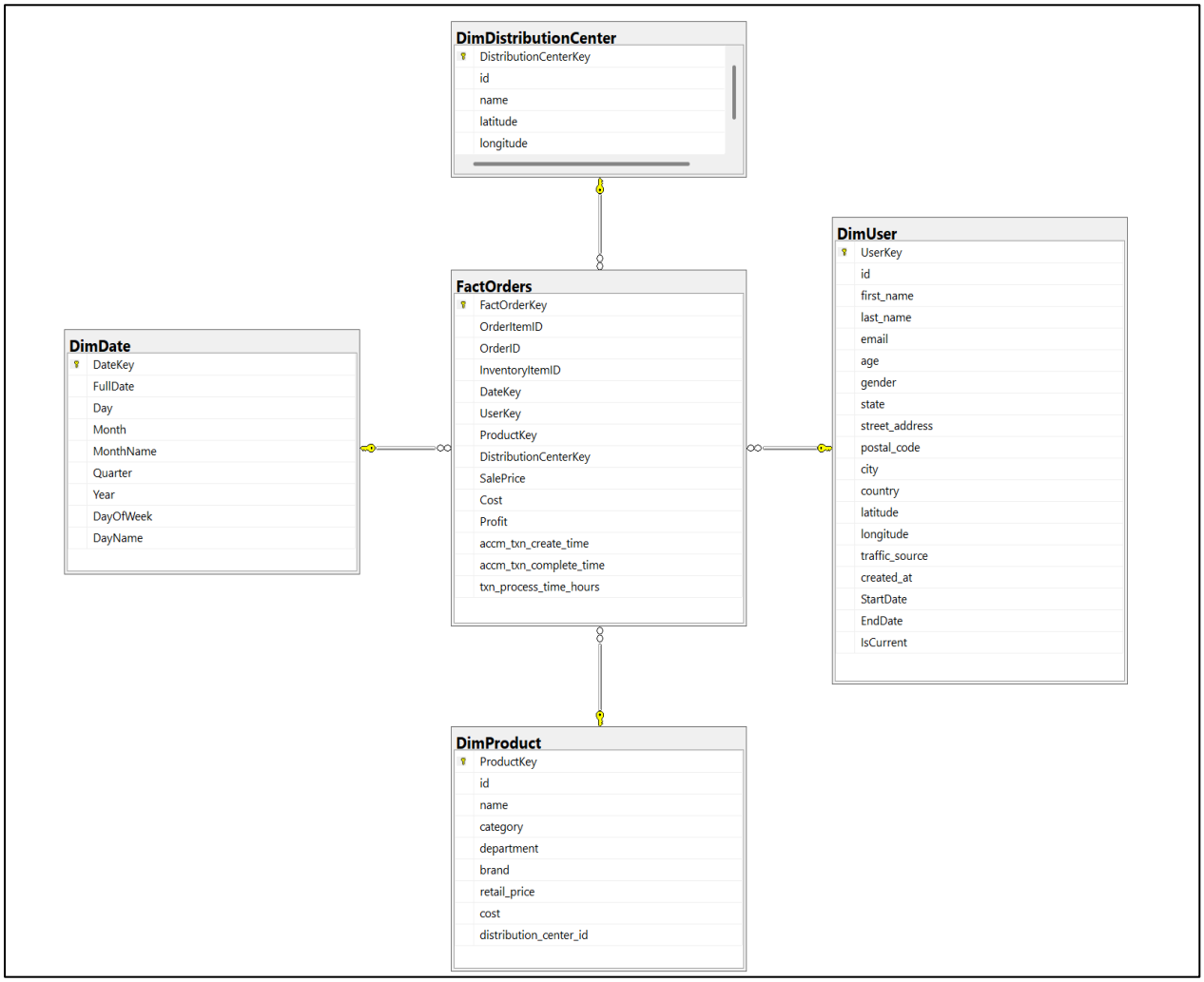
  1. SalePrice
  2. Cost
  3. Profit
  4. Transaction Process Time

Dimension Tables:

- **DimUser -** Stores customer details (name, gender, location) for customer-based analysis with SCD Type 2 tracking.

- **DimProduct** - Stores product details for analyzing product performance.
- **DimDistributionCenter** - Stores distribution center details for logistics analysis.
- **DimDate** - Stores date information for time-based analysis.

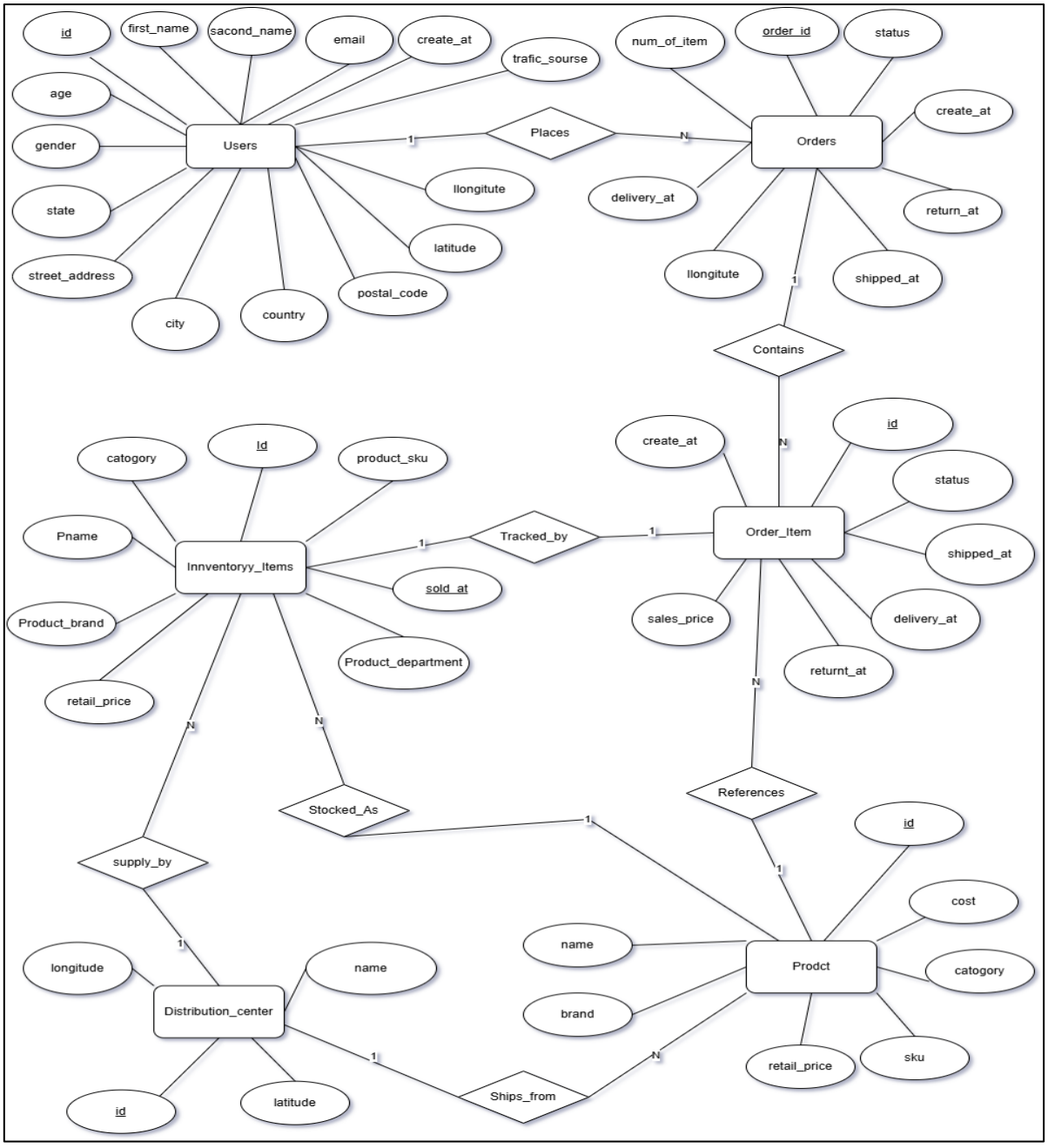
The fact table is connected to all dimensions using foreign keys.



To support a realistic ETL process and effectively demonstrate the integration of multiple heterogeneous data sources using SSIS, the original single Excel dataset was logically divided into three distinct source types. This separation was performed based on the entities and relationships identified in the ER diagram, ensuring proper data organization, improved data handling, and alignment with real-world data warehousing practices.

Conceptual ER Diagram

The following ER diagram illustrates the conceptual data model of the original E-commerce data source. This source data was transformed via ETL (SSIS) in Assignment 1 into the star schema used in EDW.

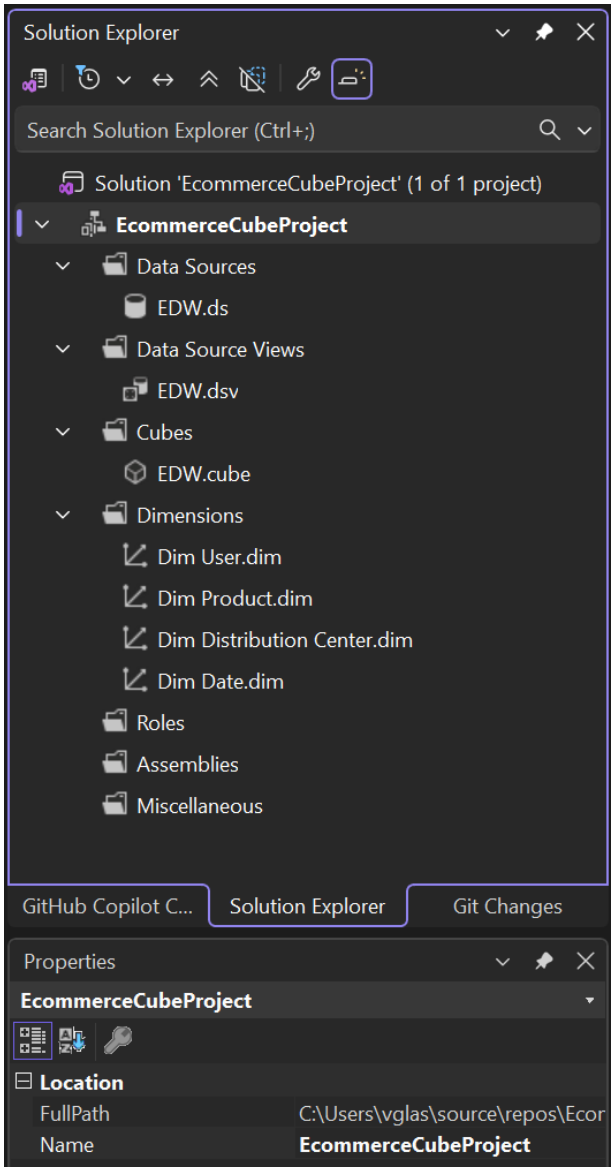


SSAS Cube Implementation

Creating the SSAS Project:

A new Analysis Services Multidimensional project was created in Visual Studio.

After creation, the Solution Explorer shows the standard SSAS project structure. This structure includes folders for Data Sources, Data Source Views, Cubes, Dimensions, Roles, Assemblies, and Miscellaneous.

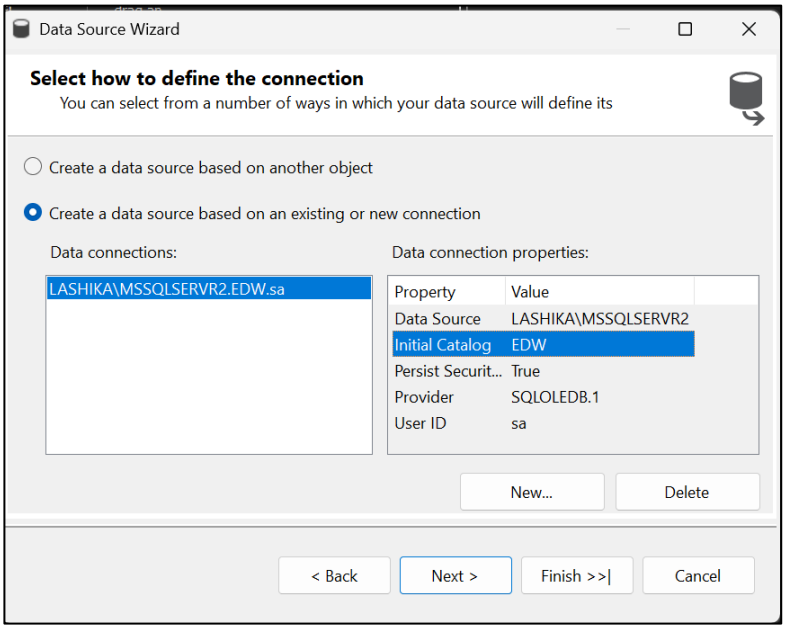


Solution Explorer showing project structure

Creating the Data Source (DS_EDW):

A data source was created to connect the SSAS project to the EDW database.

1. Right-clicked Data Sources in Solution Explorer and selected New Data Source
2. Created a new connection
3. Named the data source DS_EDW
4. Set impersonation to Use the service account



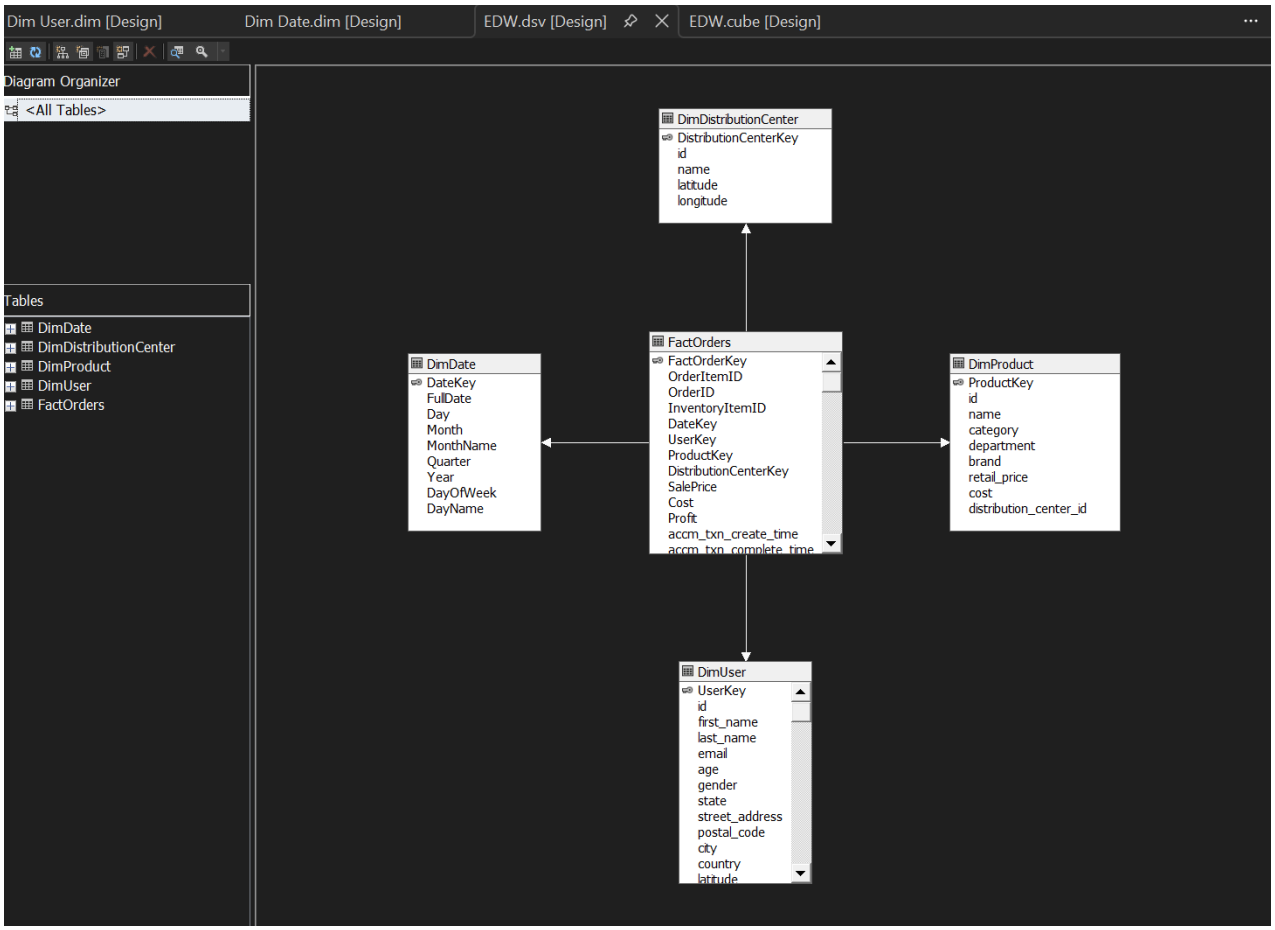
Data Source Configuration

Creating the Data Source View (DSV_EDW):

A Data Source View was created to define the logical schema.

1. Right-click Data Source Views → New Data Source View
2. Select DS_EDW as the relational data source
3. Add all five tables from the EDW database

The DSV designer automatically detected the foreign key relationships between the tables. The following relationships were established



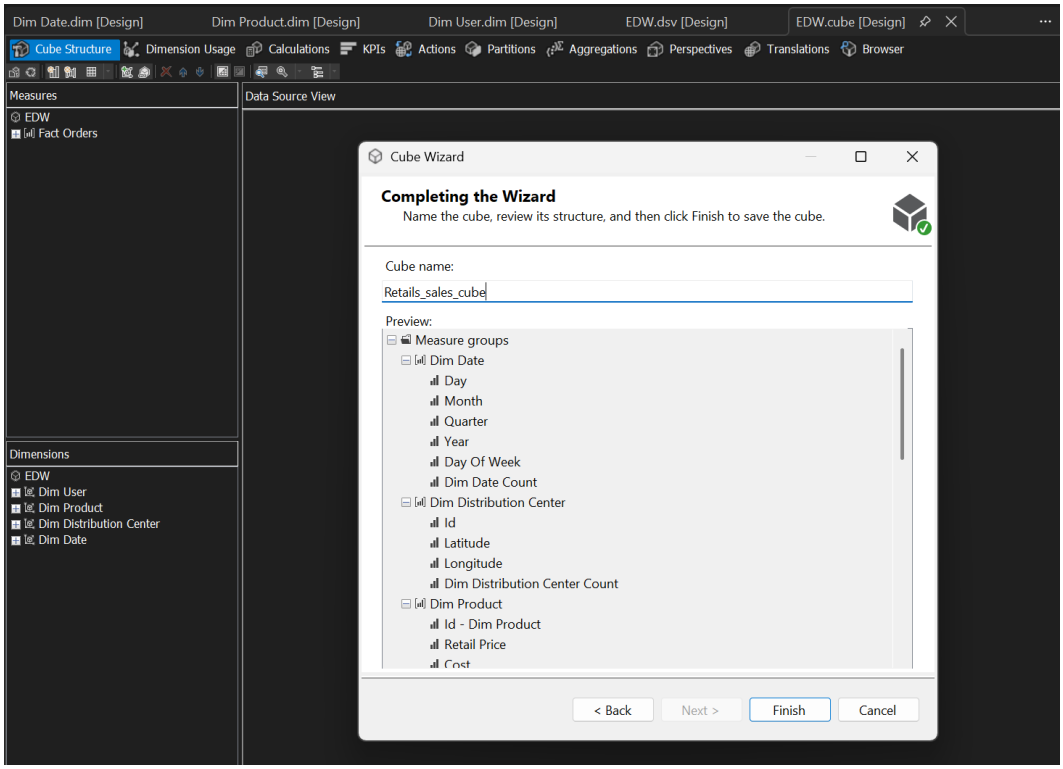
DSV with Relationships

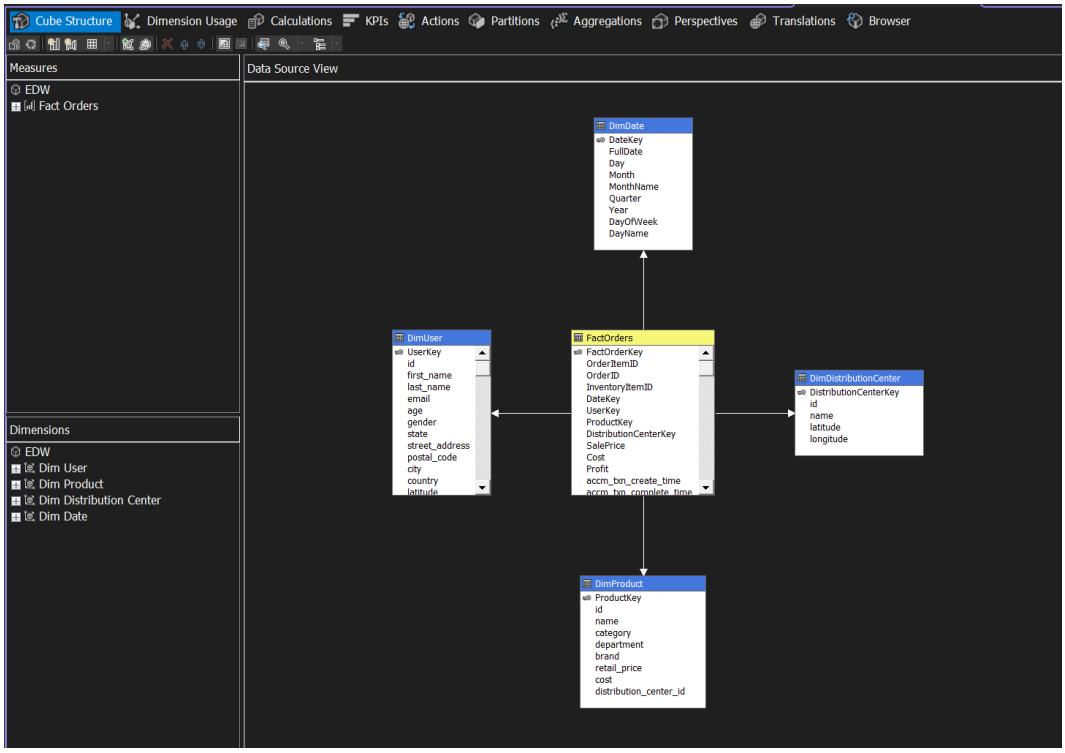
Creating the Cube (Cube_EDW):

The cube was created using the Cube Wizard

1. Right-clicked Cubes and selected New Cube
2. Selected Use existing tables as the creation method.
3. Selected FactSales as the measure group table.
4. Selected the following measures from FactSales (e.g., TotalSales)
5. All five-dimension tables were automatically detected and included
6. Named the cube as Cube_EDW

The cube allows for effective multidimensional analysis using defined hierarchies and relationships (eg: Sales by Region, Product Category, Time periods, Customer Segment, etc.).


Dim Date.dim [Design]
Dim Product.dim [Design]
Dim User.dim [Design]
EDW.dsv [Design]
EDW.cube [Design]
Cube Structure
Dimension Usage
Calculations
KPIs
Actions
Partitions
Aggregations
Perspectives
Translations
Browser
Measures
EDW
Fact Orders
Data Source View
Cube Wizard
Completing the Wizard
Name the cube, review its structure, and then click Finish to save the cube.
Cube name:
Retails_sales_cube
Preview:
Measure groups
Dim Date
Day
Month
Quarter
Year
Day Of Week
Dim Date Count
Dim Distribution Center
Id
Latitude
Longitude
Dim Distribution Center Count
Dim Product
Id - Dim Product
Retail Price
Cost
Dimensions
EDW
Dim User
Dim Product
Dim Distribution Center
Dim Date
< Back
Next >
Finish
Cancel

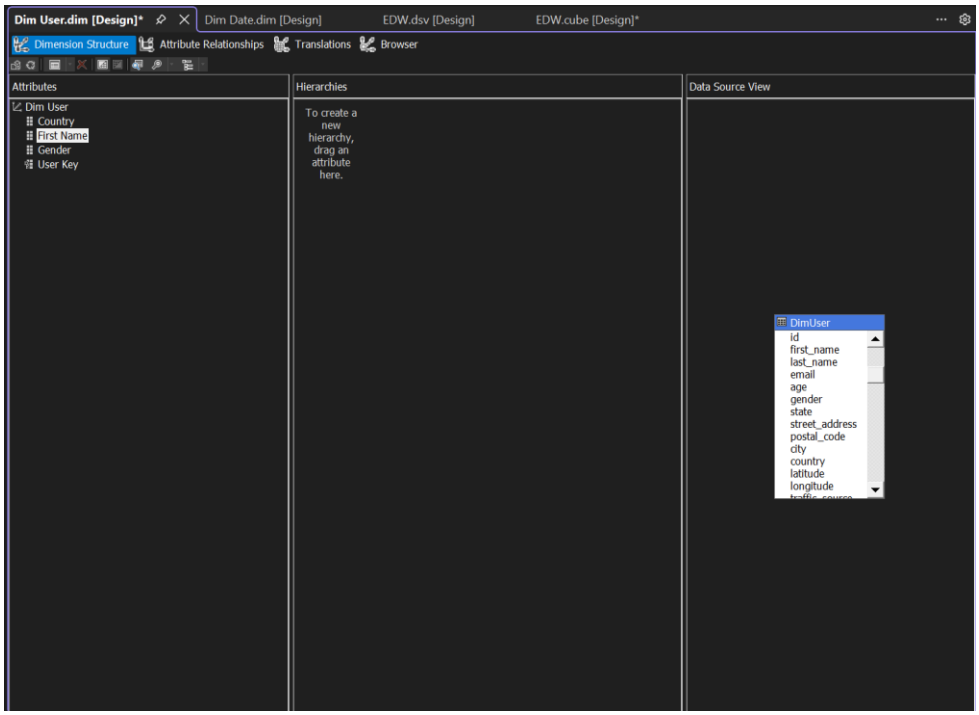

Cube Structure
Dimension Usage
Calculations
KPIs
Actions
Partitions
Aggregations
Perspectives
Translations
Browser
Measures
EDW
Fact Orders
Data Source View
Dimensions
EDW
Dim User
Dim Product
Dim Distribution Center
Dim Date
DimDate
DateKey
FullDate
Day
Month
MonthName
Quarter
Year
DayOfWeek
DayName
DimUser
UserKey
id
first_name
last_name
email
age
gender
state
street_address
postal_code
city
country
FactOrders
FactOrderKey
OrderItemID
OrderID
InventoryItemID
DateKey
UserKey
ProductKey
DistributionCenterKey
SalePrice
Cost
Profit
DimDistributionCenter
DistributionCenterKey
id
name
latitude
longitude
DimProduct
ProductKey
id
name
category
department
brand
retail_price
cost
distribution_center_id

Adding Attributes to Dimensions:

Each dimension was opened in the Dimension Designer and relevant attributes were added by dragging columns from the Data Source View pane to the Attributes pane.

DimUser:

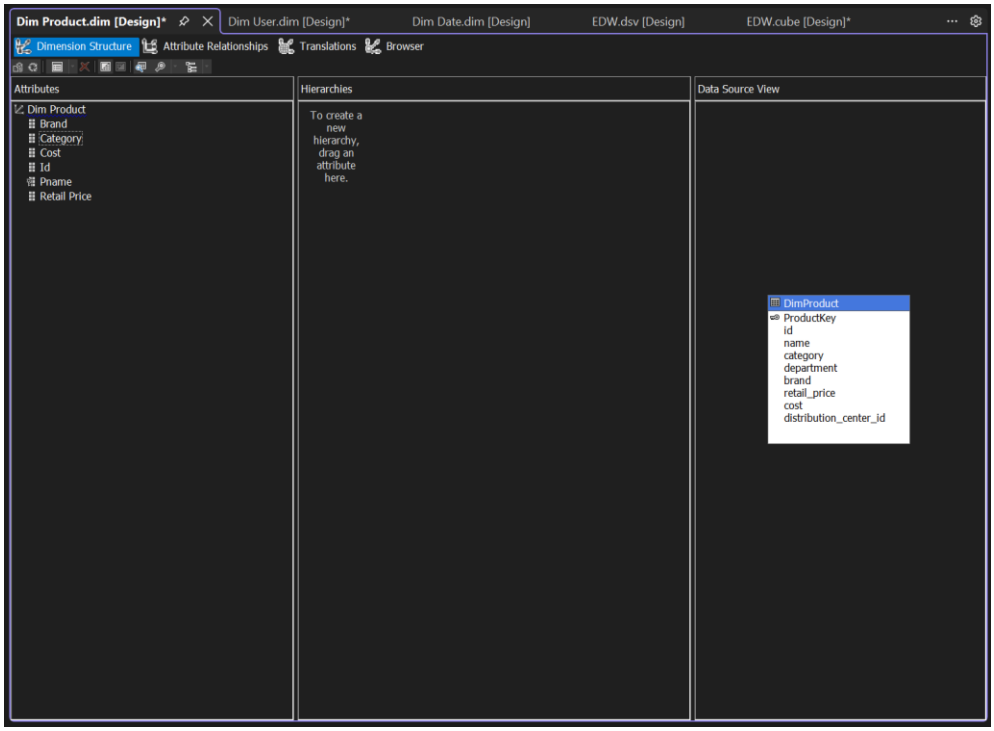
Attributes added: AlternateUserID (User key), Name, Gender, Country



DimUser dimension with attributes

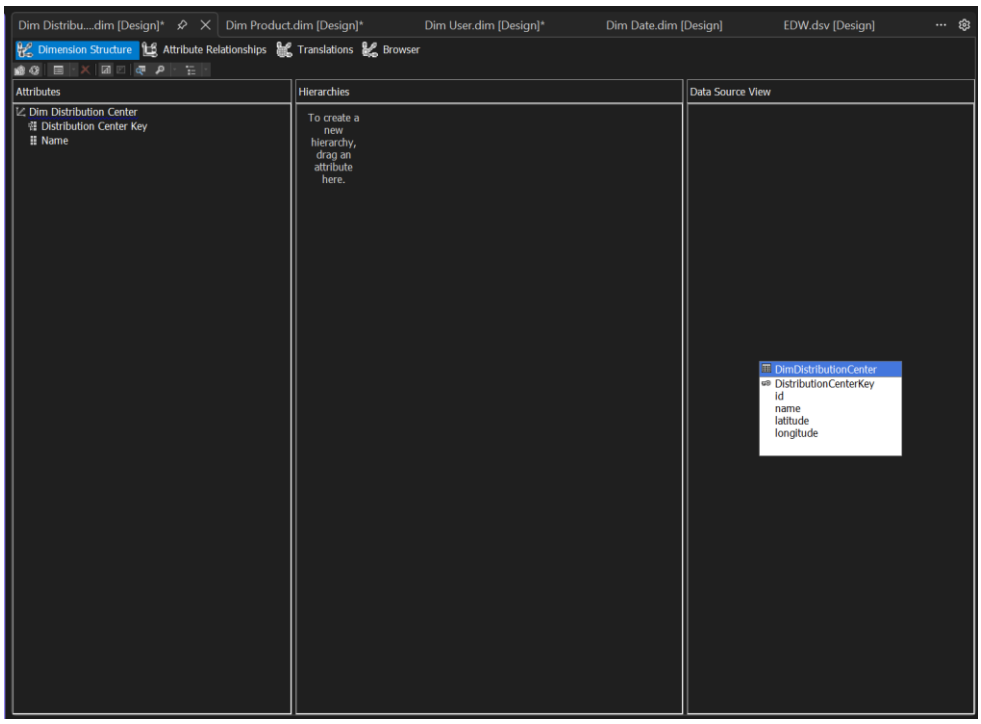
DimProduct:

Attributes added: Category, Product ID, Product Name, Brand, Retail Price, Cost.



DimDistributionCenter:

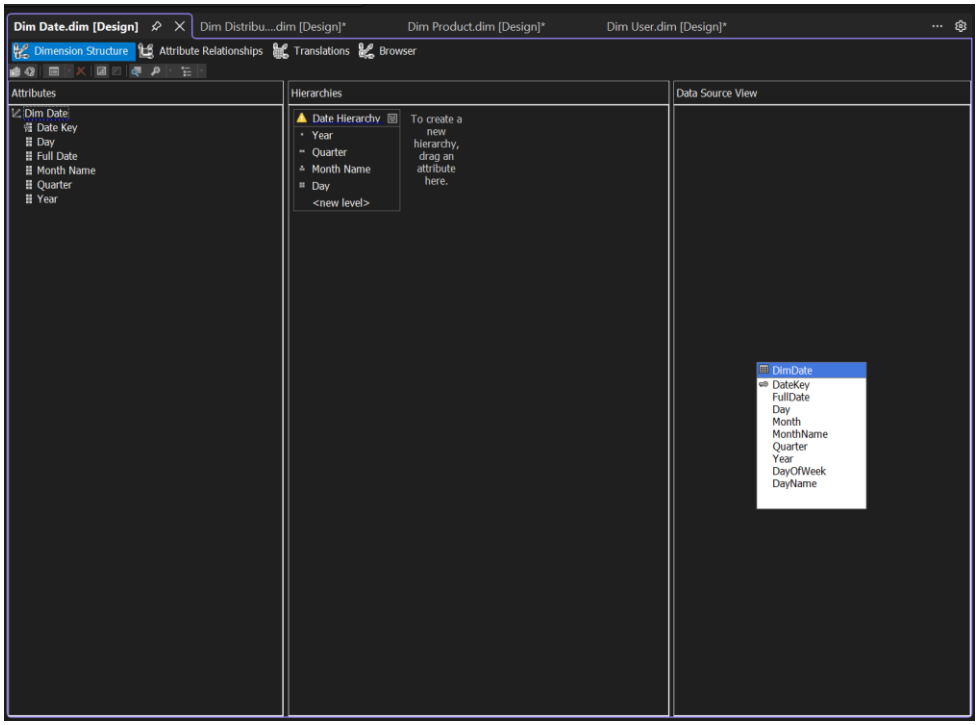
Attributes added: AlternateUserID (DistributionCenterKey), Name



DimDistributionCenter dimension with attributes

DimDate:

Attributes added: Date Key, Full Date, Day, Month Name, Quarter, Year



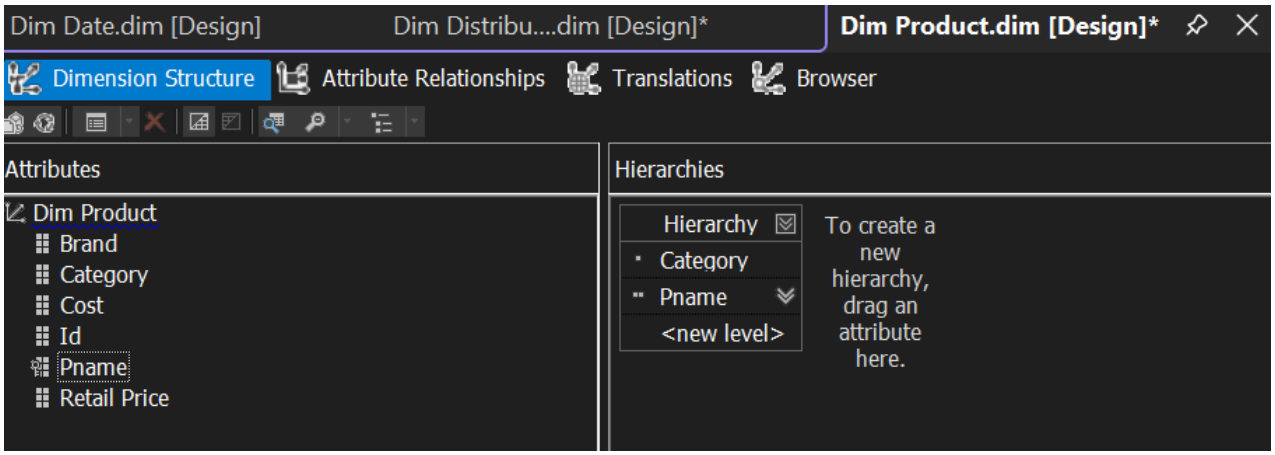
Creating Hierarchies:

Two hierarchies were created to enable drill-down navigation in the cube

1. **Product Hierarchy (DimProduct):**

The DimProduct dimension has a defined hierarchy. This hierarchy has two levels: Category (the Top Level) and Product Name (the Lowest Level). Users will be able to navigate from a high level of a Category to a detailed level of Product Name within this structure.

- Category → Product Name

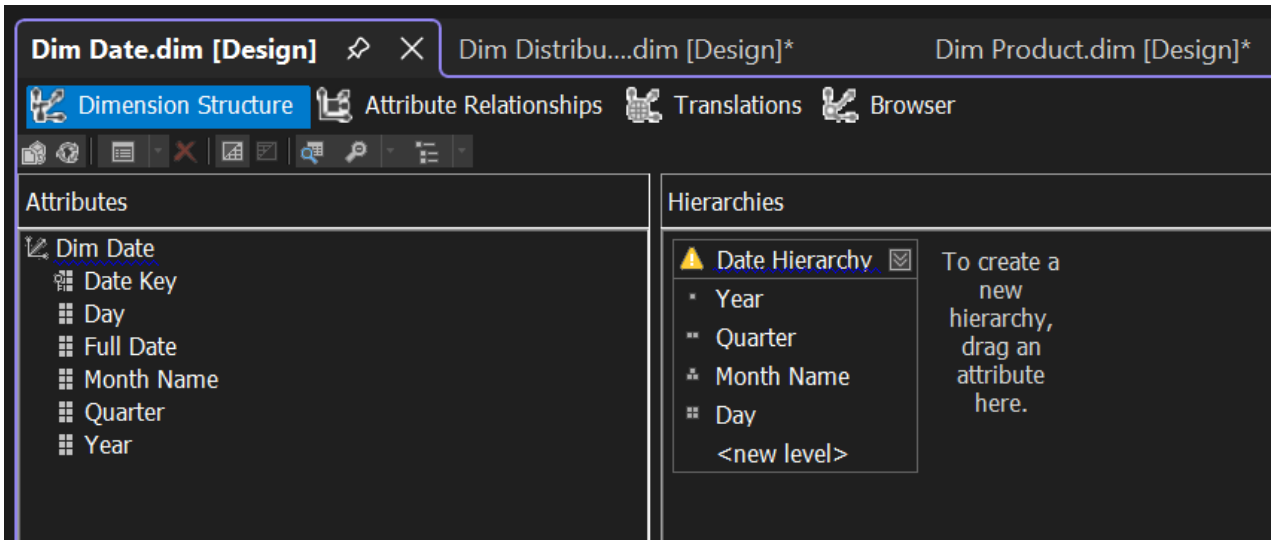


2. **Date Hierarchy:**

A hierarchy was created on the DimDate dimension with four levels

- Year → Quarter → Month Name → Day

This hierarchy allows users to drill down from yearly aggregates to daily detail.



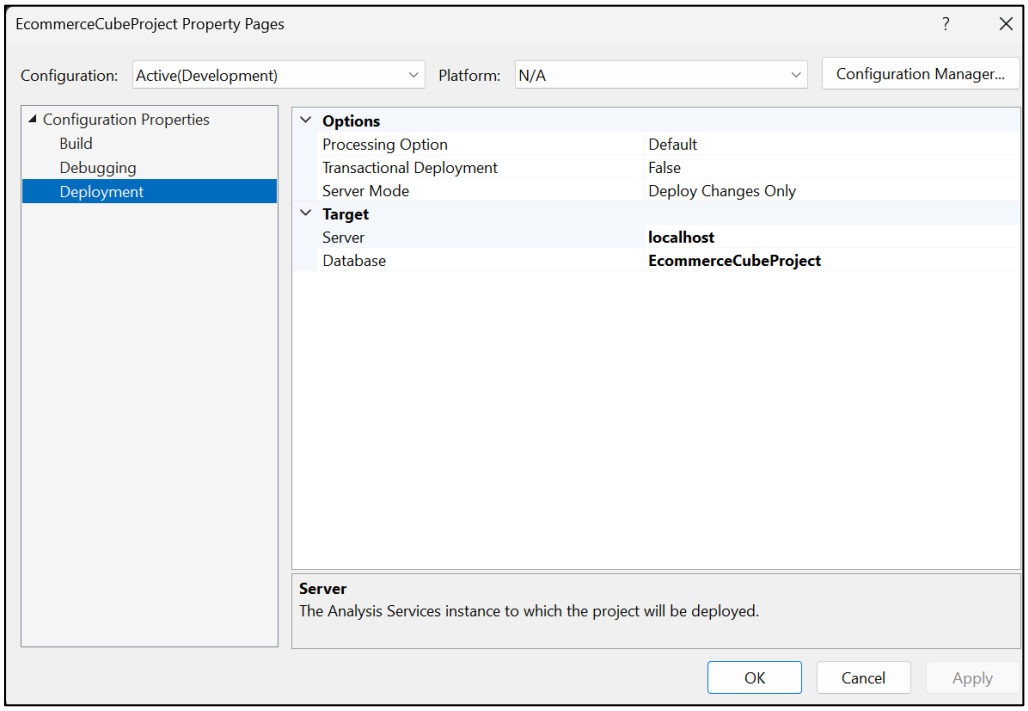
These hierarchies allow users to perform drill-down and roll-up operations efficiently.

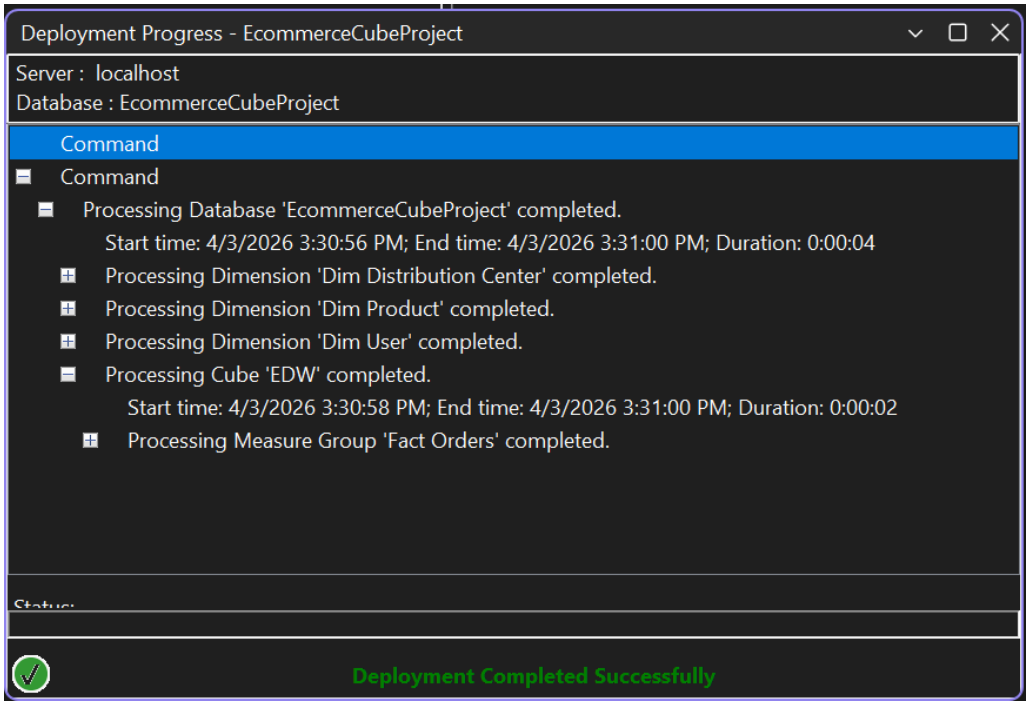
Deploying the Cube:

The cube was deployed to the SSAS server and processed the cube to populate data for analysis.

3. Right-click project → Properties
4. Set deployment server
5. Click Deploy

6. Process cube successfully





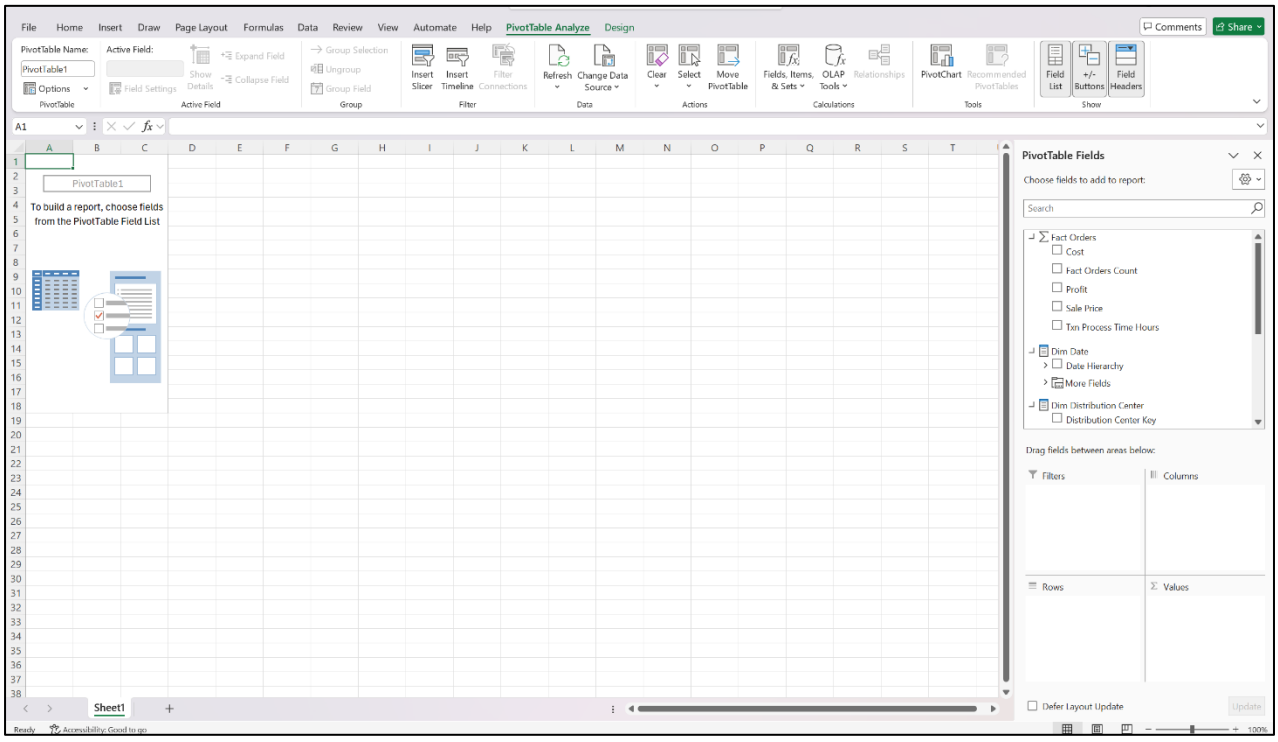
Confirms successful deployment and processing of cube.

Demonstration of OLAP operations

To analyze data, we connected Microsoft Excel to the SSAS cube. Here are the steps taken to establish the connection:

1. Opened Microsoft Excel and navigated to Data → From Database → From Analysis Services.
2. In the Data Connection Wizard, entered the server name LASHIKA\MSSQLSERVR2.
3. Initially attempted to use SQL authentication but was unsuccessful. Changed the authentication type to Windows Authentication. Successfully connected.
4. Selected database EcommerceCubeProject and cube EDW_Cube.

After completing these steps, the PivotTable Field List displayed all the measures and dimensions such as users, products, dates, and order-related data. This allowed easy analysis using filtering, summarizing, and drill-down operations.

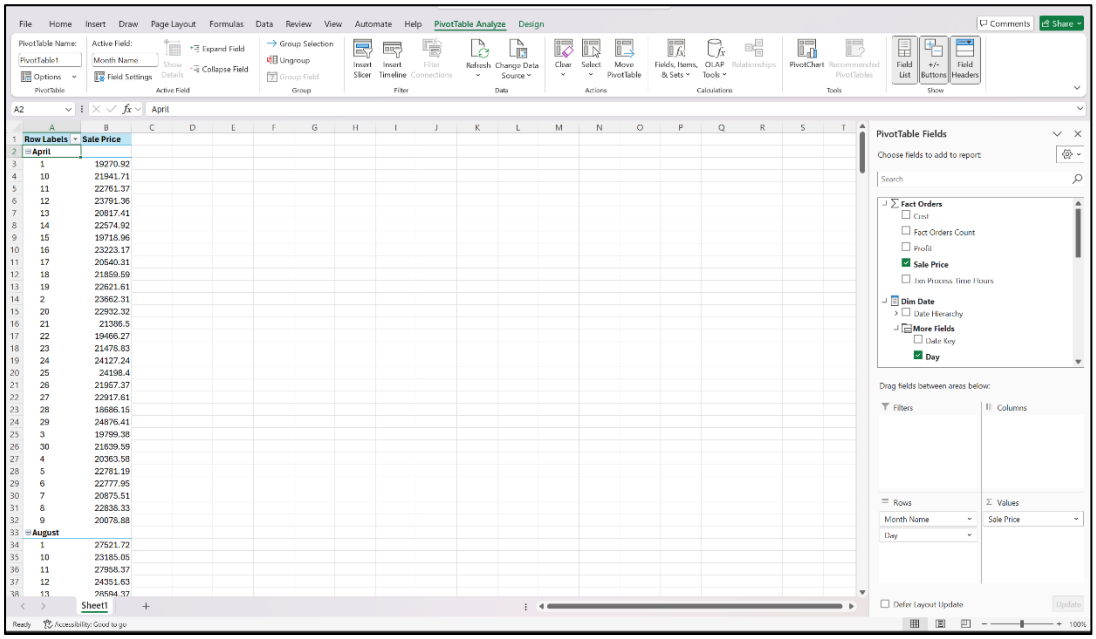


1.Roll-up / Drill-down:

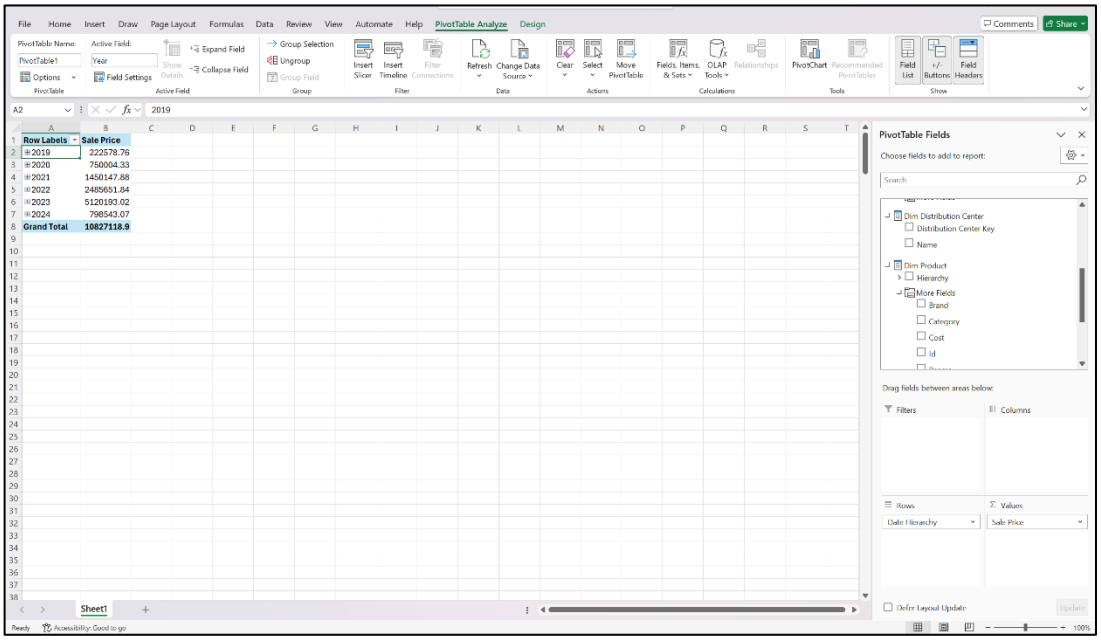
- Roll-up: Collapse detail to see higher-level summary (e.g., see total Sales by Year).
- Drill-down: Expand to see more detail (eg: see Months or Quarters under a specific Year).

Process:

Click the "-" next to the month to roll up and collapse the details.



Click the "+" next to a year (eg: 2019) to drill down and see months.



2. Slice:

- Filtering the data by a single dimension (eg: only show results for one Customer Segment).

Process:

1. Dragged a field (eg: Country) to the Filters area.
2. At the top of my PivotTable, used the drop-down to select the value that wanted.

3. The PivotTable now only shows data for the selected Segment.

	A	B	C
1	Pname	Low Profile Dyed Cotton Twill Cap - Navy W39S55D	
2			
3	**Row Labels**	**Sale Price**	
4	**2022**		
5	**2**		
6	April		
7	2	6.25	
8	20	6.25	
9	**3**		
10	July		
11	18	6.25	
12	**2023**	**6.25**	
13	**Grand Total**	**25**	
14			

3. Dice:

- Filtering the data by multiple dimensions at once (eg: Country, Product name, and Year).

Process:

1. Dragged multiple fields (Country, Product name, and Year) and added them as Slicers.
2. Selected specific values from each slicer.

• The PivotTable now shows data for the combination of the selected dimensions.

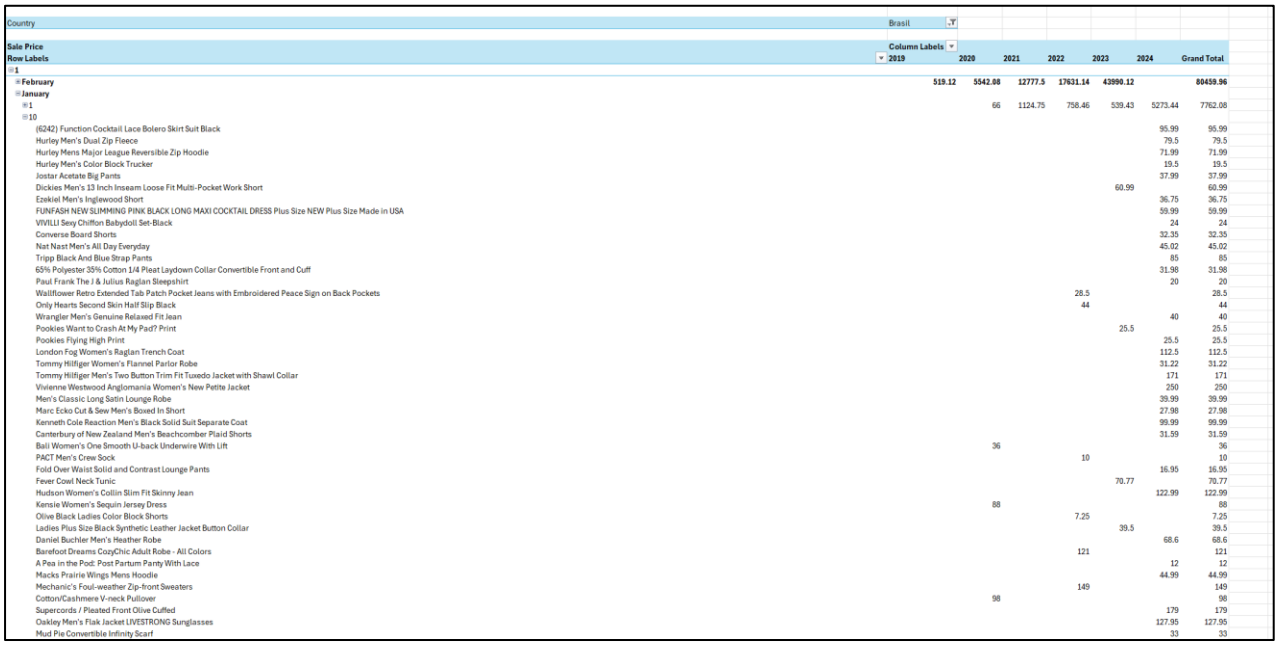
	A	B	C
1	Pname	Washed Canvas Ivy Cap - Black W11S64C	
2	Country	Brasil	
3	Year	2020	
4			
5	**Row Labels**	**Sale Price**	
6	**2**		
7	**April**		
8	21	15.99	
9	**Grand Total**	**15.99**	
10			

4. Pivot:

- Swapping rows and columns to view data from different perspectives.

Process:

1. Dragged the Product Category Hierarchy from the field list into the Rows area.
2. Dragged the Date Hierarchy from the field list into the Columns area.
3. Dragged a field from Columns to Rows, or vice versa, to change the layout.



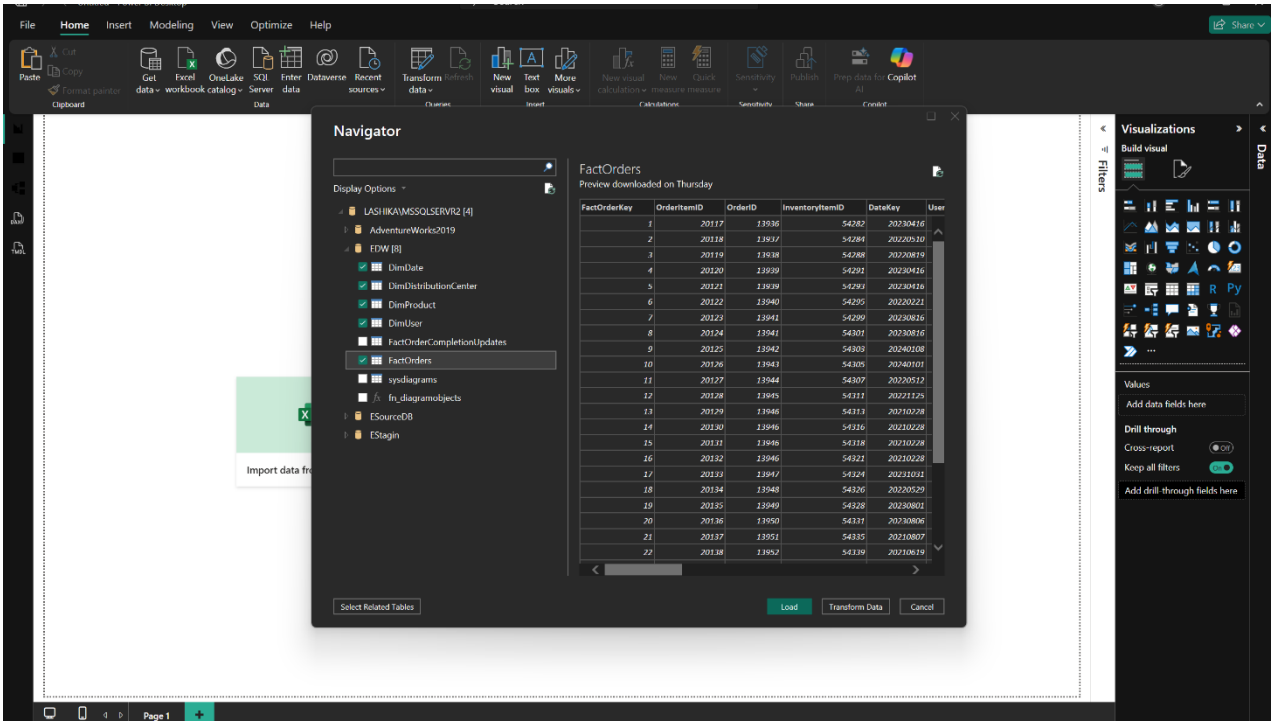
Country	Brasil						
Sale Price	Column Labels						
Row Labels	2019	2020	2021	2022	2023	2024	Grand Total
1							
February	519.12	5542.08	12777.5	17631.14	43990.12		80459.96
January							
1		66	1124.75	758.46	539.43	5273.44	7762.08
10							
(6242) Function Cocktail Lace Bolero Skirt Suit Black						95.99	95.99
Hurley Men's Dual Zip Fleece						79.5	79.5
Hurley Mens Major League Reversible Zip Hoodie						71.99	71.99
Hurley Men's Color Block Trucker						19.5	19.5
Jostar Acetate Big Pants						37.99	37.99
Dickies Men's 13 Inch Inseam Loose Fit Multi-Pocket Work Short					60.99		60.99
Ezekiel Men's Inglewood Short						36.75	36.75
FUNFASH NEW SLIMMING PINK BLACK LONG MAXI COCKTAIL DRESS Plus Size NEW Plus Size Made in USA						59.99	59.99
VIVILLI Sexy Chiffon Babydoll Set-Black						24	24
Converse Board Shorts						32.35	32.35
Nat Nast Men's All Day Everyday						45.02	45.02
Tripp Black And Blue Strap Pants						85	85
65% Polyester 35% Cotton 1/4 Pleat Laydown Collar Convertible Front and Cuff						31.98	31.98
Paul Frank The J & Julius Raglan Sleepshirt						20	20
Wallflower Retro Extended Tab Patch Pocket Jeans with Embroidered Peace Sign on Back Pockets				28.5			28.5
Only Hearts Second Skin Half Slip Black				44			44
Wrangler Men's Genuine Relaxed Fit Jean						40	40
Pookies Want to Crash At My Pad? Print					25.5		25.5
Pookies Flying High Print						25.5	25.5
London Fog Women's Raglan Trench Coat						112.5	112.5
Tommy Hilfiger Women's Flannel Parlor Robe						31.22	31.22
Tommy Hilfiger Men's Two Button Trim Fit Tuxedo Jacket with Shawl Collar						171	171
Vivienne Westwood Anglomania Women's New Petite Jacket						250	250
Men's Classic Long Satin Lounge Robe						39.99	39.99
Marc Ecko Cut & Sew Men's Boxed In Short						27.98	27.98
Kenneth Cole Reaction Men's Black Solid Suit Separate Coat						99.99	99.99
Canterbury of New Zealand Men's Beachcomber Plaid Shorts						31.59	31.59
Bali Women's One Smooth U-back Underwire With Lift		36					36
PACT Men's Crew Sock				10			10
Fold Over Waist Solid and Contrast Lounge Pants						16.95	16.95
Fever Cowl Neck Tunic					70.77		70.77
Hudson Women's Collin Slim Fit Skinny Jean						122.99	122.99
Kensie Women's Sequin Jersey Dress		88					88
Olive Black Ladies Color Block Shorts				7.25			7.25
Ladies Plus Size Black Synthetic Leather Jacket Button Collar					39.5		39.5
Daniel Buchler Men's Heather Robe						68.6	68.6
Barefoot Dreams CozyChic Adult Robe - All Colors				121			121
A Pea in the Pod: Post Partum Panty With Lace						12	12
Macks Prairie Wings Mens Hoodie						44.99	44.99
Mechanic's Foul-weather Zip-front Sweaters				149			149
Cotton/Cashmere V-neck Pullover		98					98
Supercords / Pleated Front Olive Cuffed						179	179
Oakley Men's Flak Jacket LIVESTRONG Sunglasses						127.95	127.95
Mud Pie Convertible Infinity Scarf						33	33

Power BI Reports:

Prepare Data in Power BI:

Opened the Power BI Desktop then clicked Import data from SQL server, entered my server and database then loaded my fact and dimension tables. Data Sources

- Source: SQL Server database (EcommerceDB).



Creating Required Reports

Report 1: Matrix visual

• Created a Matrix visual to display detailed tabular data with row and column groupings (Customer_Name grouped by Year, showing aggregated Sales, Profit, and Quantity with subtotals and grand totals).

Sales by Product and Year

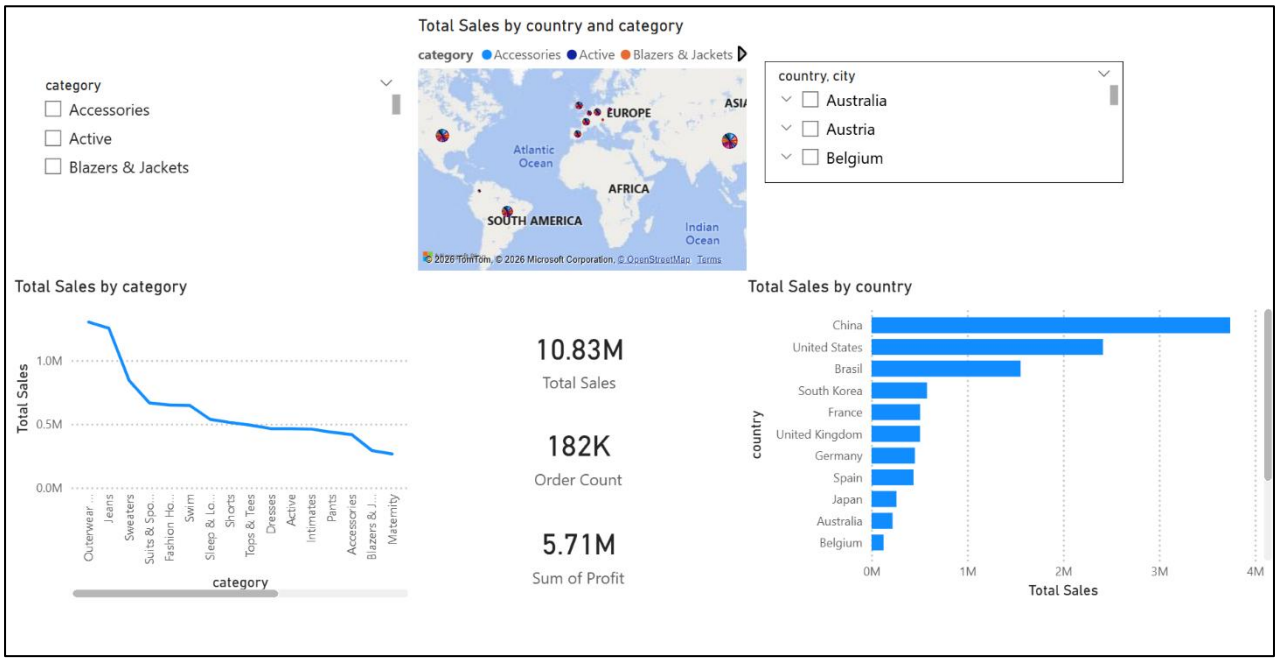
Year / name	ount	2022 Total Sales	2022 Order Count	2023 Total Sales	2023 Order Count	2024 Total Sales	2024 Order Count	Total Total Sales	Total Order Count
	1	147.99	1	551.97	6	147.99	1	919.95	10
!iT Jeans Junior's Billy Shorts	2	96.00	2			48.00	1	336.00	7
!iT Jeans Women's Bonaroo Slouch Shorts		22.00	2					33.00	3
!iT Jeans Women's Bureau Trouser Denim Short		36.29	1	36.29	1			108.87	3
!iT Jeans Women's Coachella Canyon Stretch Shorts		25.00	1					25.00	1
!iT Jeans Women's Coachella Palooza Shorts				33.00	3			33.00	3
!iT Jeans Women's Coachella Shorts	1			175.00	7			250.00	10
!iT Jeans Women's Curvy Slim Boot Jean	3	261.00	3	783.00	9	87.00	1	1,566.00	18
!iT Jeans Women's Curvy Stiletto Jean	1	211.85	3	555.56	8			819.28	12
!iT Jeans Women's Harvest Crop Jean		86.13	3	86.13	3			172.26	6
!iT Jeans Women's Skinny Micro Printed Corduroy	1	111.72	2					279.30	5
#5098 Satin V-neck Bridemaid Prom Formal Long Gown	1	178.00	2	267.00	3	89.00	1	801.00	9
#7701 Halter Corset Bodice Bridesmaids Evening Formal Dress		109.00	1	218.00	2			436.00	4
#7716 One Shoulder Chiffon Homecoming Bridesmaids Party Dress		89.00	1	89.00	1			178.00	2
(1272) Ladies Wide Leg Smart City Trousers Light Brown		55.98	2	55.98	2			111.96	4
(1272) Wide Leg Smart Soft City Trousers Black	1	55.98	2	111.96	4			195.93	7
(1279) Skinny Trousers Military Army Look Camouflage	1	24.99	1					49.98	2
(2328) Satin Pencil Skirt with Skinny Belt Purple	2	71.97	3	71.97	3			191.92	8
(2382) Waist Detail Pencil Skirt Shadow Stripe	1			24.99	1			49.98	2
(2384) Fishtail Flute Skirt with Free Skinny Belt Beige		24.99	1	24.99	1	24.99	1	74.97	3
(2384) Fishtail Peplum Hem Pencil Skirt Black				74.97	3			99.96	4
(6239) Waist Detail Pencil Skirt Suit Shadow Stripe Black		55.99	1			55.99	1	111.98	2
(6240) Dip Hemmed Flute Skirt Suit Pinstripe Black		127.98	2	191.97	3	127.98	2	447.93	7
(6242) Function Cocktail Lace Bolero Skirt Suit Black	1	191.98	2	191.98	2	191.98	2	671.93	7
(6249-2) Smart Satin Evening Suit with Flute Skirt Beige	2	75.99	1	75.99	1			303.96	4
(6249-3) Teddy Style Function Suit with Pencil Skirt Grey	1	151.98	2	151.98	2	75.99	1	455.94	6
(6251) Smart Casual Tailored Trouser Suit Dark Grey				167.97	3			167.97	3
(6252) Pinstripe Tailored Boot Leg Trouser Suit	1	57.99	1	173.97	3			347.94	6
(CH133NL) A Medium Full Bridal Petticoat Crinoline Slip Skirt Sz Large	3	89.97	3	179.94	6	29.99	1	389.87	13
(CH133NM) A Medium Full Bridal Petticoat Crinoline Slip Skirt Sz Medium	1	59.80	2	89.70	3			179.40	6
(ONE) 1 Satin Headband	1	6.99	1	6.99	1	6.99	1	41.94	6
(One) CHEER Rhinestone Studded Stretch Headband	1	5.99	1	23.96	4			35.94	6
(Set of 3) Leopard Animal Print Stretch Headband		18.00	2	18.00	2			36.00	4
(Set of 4) 2.5 Glitter Cotton Stretch Headbands	1	21.98	2	32.97	3	10.99	1	76.93	7
?Â/Â¿Long Sleeve Striped V-Neck Hoodie Pullover		97.00	6	131.00	8	19.50	1	247.50	15
[BLANKNYC] Women's Happy Trail Denim Shorts				115.83	3			231.66	6
Total	**314**	**2,485,651.84**	**42308**	**5,120,193.02**	**85517**	**798,543.07**	**13235**	**10,827,118.90**	**181759**

Report 2: Cascading Slicers with Multiple Visuals:

Created an interactive dashboard page titled Report 2 – Cascading Slicers with multiple slicers implementing cascading filters and several visuals (Map, Bar chart, KPI cards, and Pie chart) to present comprehensive insights from the data.

Process:

1. In Power BI Desktop, connected to the deployed EDW_Cube.
2. Added slicers for Country, City and Category to the report canvas.
3. Configured cascading filters so that selecting a value in the Region slicer dynamically filters the available options in the Category slicer.
4. Added a Map visual to display the geographical distribution of sales across the word.
5. Created a chart visual showing Sum of Sales by city.
6. Added a Pie chart visual displaying Sum of Profit by Segment.
7. Included three Card visuals (KPIs) to highlight key metrics: Total Sales, Total Profit, and Total Quantity.



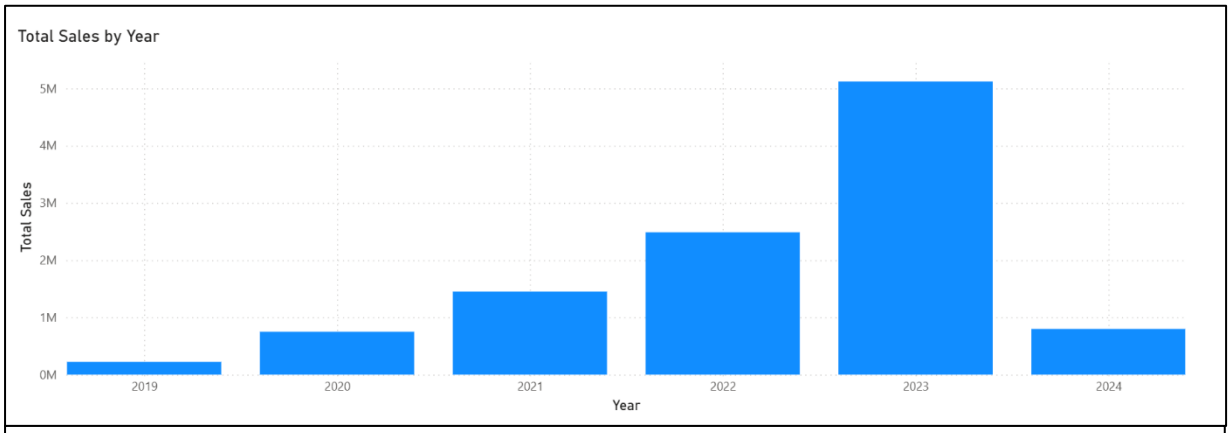
Report 3: Drill-Down Report:

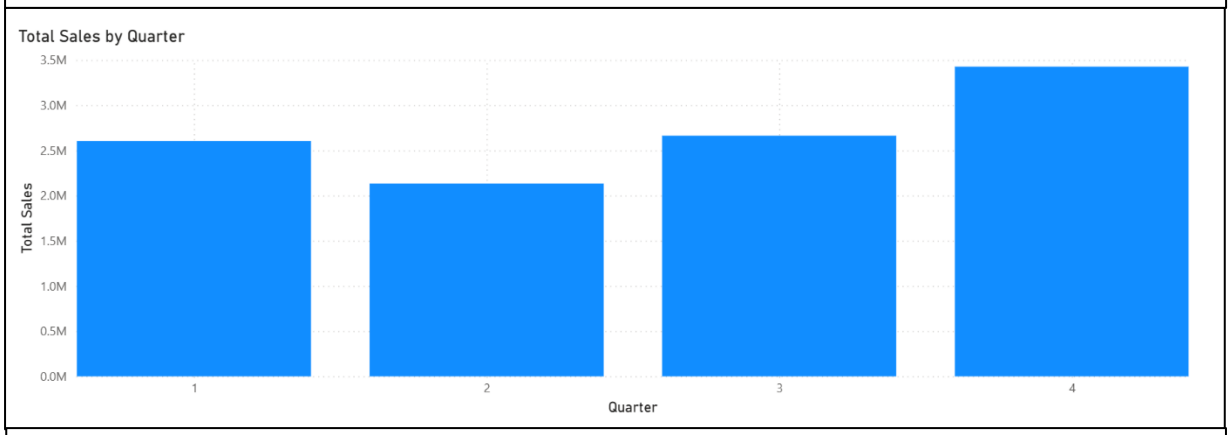
Created a drill-down report using a Column chart visual that allows users to explore sales data hierarchically from Year → Quarter → Month using the Date Hierarchy.

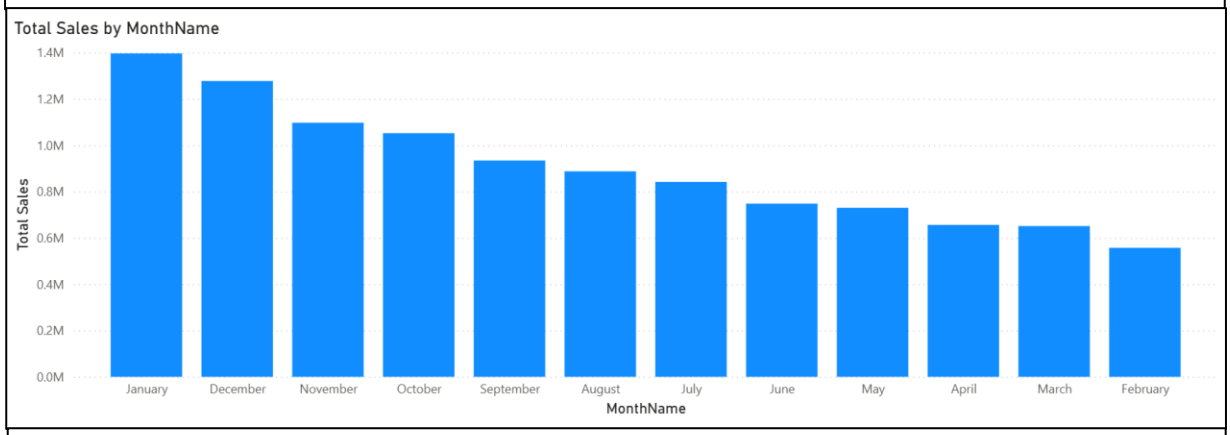
Process:

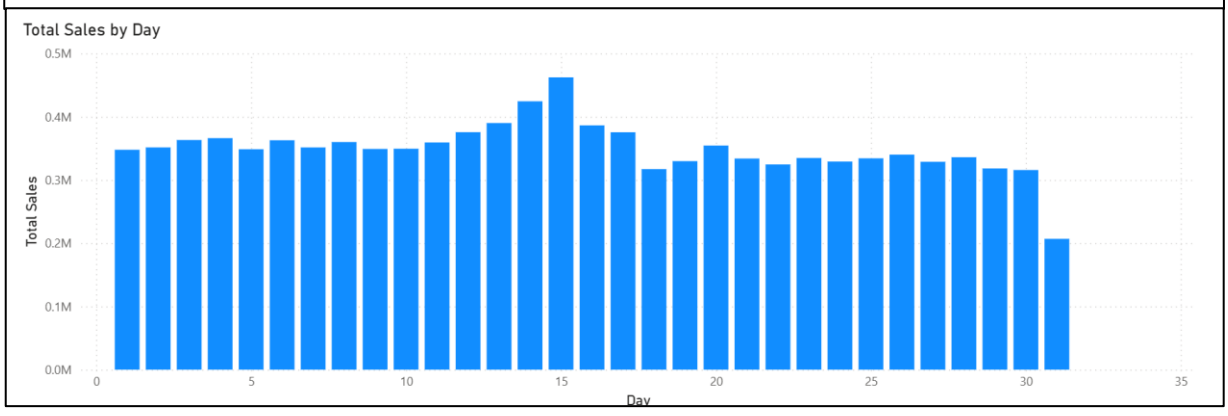
1. In Power BI Desktop, connected to the deployed EDW_Cube.
2. Created a Column chart and added Sum of Sales to the Values area.
3. Dragged the DateHierarchy (Year, Quarter, Month, day) to the Axis/Category area.

4. The chart initially displays sales aggregated by Year. Users can click the drill-down icon to expand to Quarter level, then to Month level.
5. Demonstrated all three hierarchy levels: Sales by Year, by Quarter, by Month, and by day .





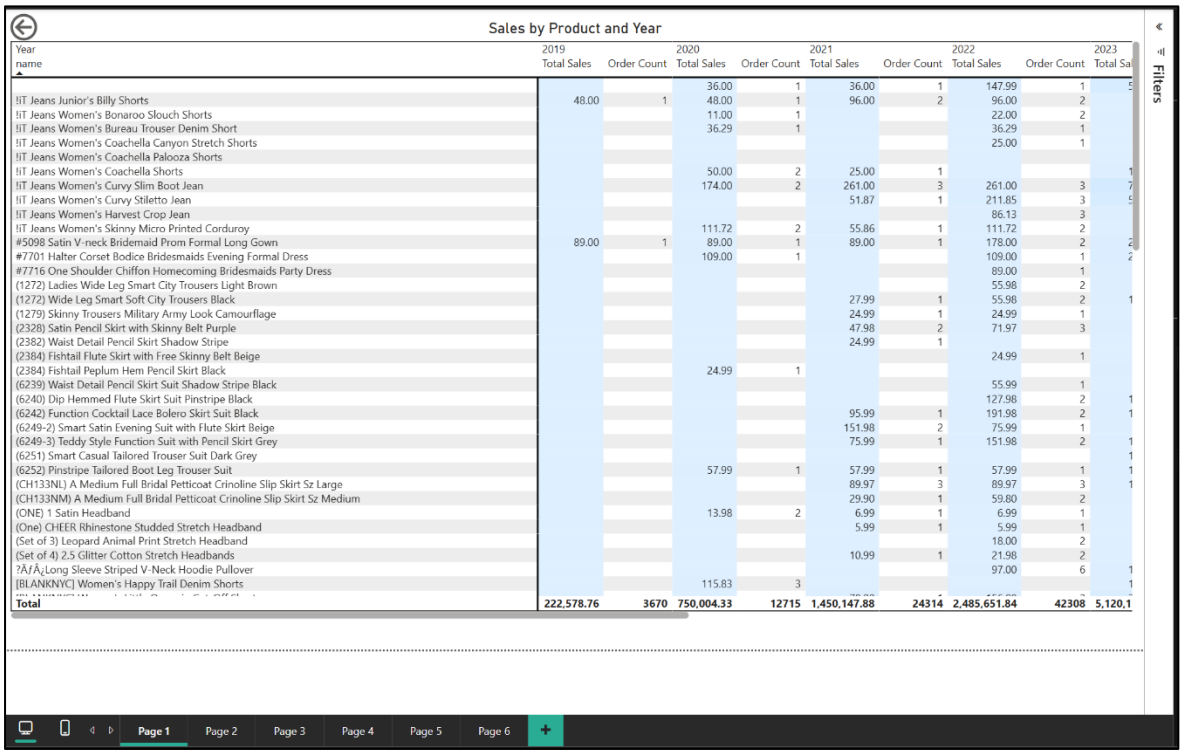




Report 4: Drill-through Report:

Created a drill-through report consisting of two pages: Report 2 -Summary (bar chart and matrix showing Sum of Sales by Category).

Report 01- detailed table. Users can right-click any bar in the visual to drill through and view related granular information (Product_Name, Sales, Order count, and Total Sales).

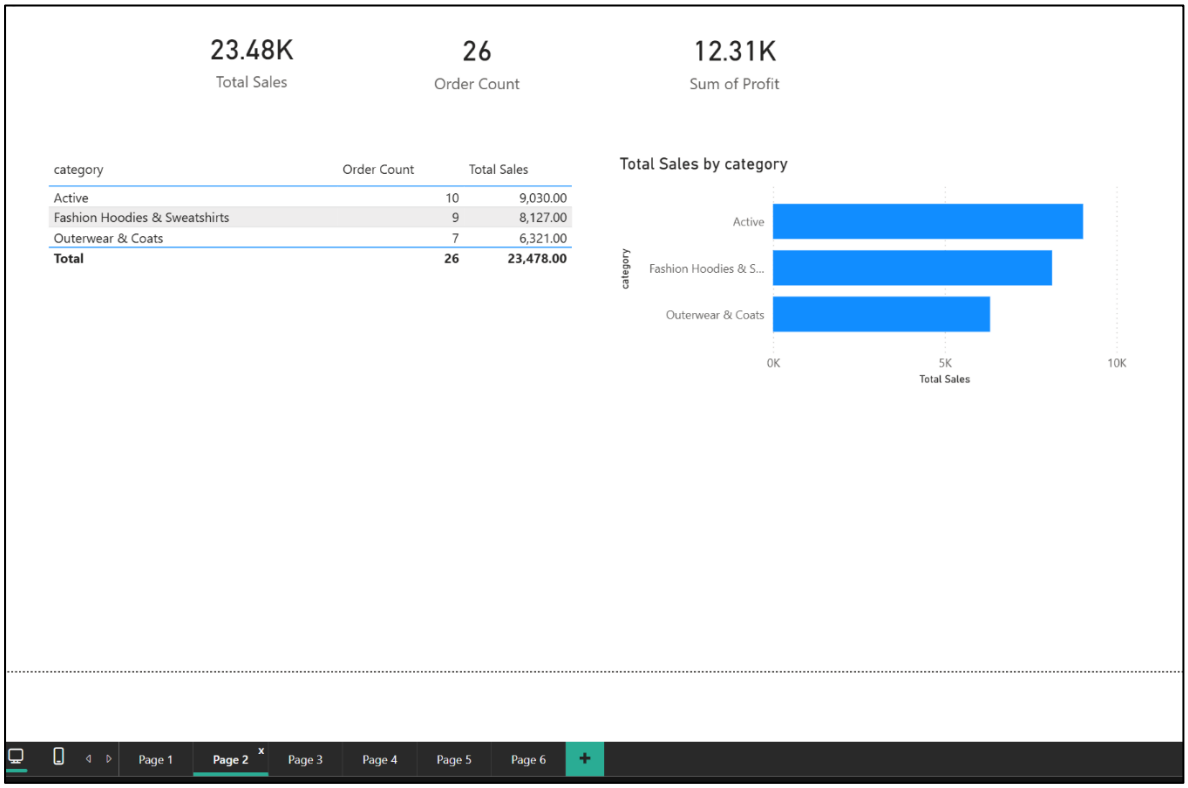


Sales by Product and Year

Year name	2019 Total Sales	Order Count	2020 Total Sales	Order Count	2021 Total Sales	Order Count	2022 Total Sales	Order Count	2023 Total Sa
			36.00	1	36.00	1	147.99	1	5
!iT Jeans Junior's Billy Shorts	48.00	1	48.00	1	96.00	2	96.00	2	
!iT Jeans Women's Bonaroo Slouch Shorts			11.00	1			22.00	2	
!iT Jeans Women's Bureau Trouser Denim Short			36.29	1			36.29	1	
!iT Jeans Women's Coachella Canyon Stretch Shorts							25.00	1	
!iT Jeans Women's Coachella Palooza Shorts									
!iT Jeans Women's Coachella Shorts			50.00	2	25.00	1			1
!iT Jeans Women's Curvy Slim Boot Jean			174.00	2	261.00	3	261.00	3	7
!iT Jeans Women's Curvy Stiletto Jean					51.87	1	211.85	3	5
!iT Jeans Women's Harvest Crop Jean							86.13	3	
!iT Jeans Women's Skinny Micro Printed Corduroy			111.72	2	55.86	1	111.72	2	
#5098 Satin V-neck Bridemaid Prom Formal Long Gown	89.00	1	89.00	1	89.00	1	178.00	2	2
#7701 Halter Corset Bodice Bridesmaids Evening Formal Dress			109.00	1			109.00	1	2
#7716 One Shoulder Chiffon Homecoming Bridesmaids Party Dress							89.00	1	
(1272) Ladies Wide Leg Smart City Trousers Light Brown							55.98	2	
(1272) Wide Leg Smart Soft City Trousers Black					27.99	1	55.98	2	1
(1279) Skinny Trousers Military Army Look Camourflage					24.99	1	24.99	1	
(2328) Satin Pencil Skirt with Skinny Belt Purple					47.98	2	71.97	3	
(2382) Waist Detail Pencil Skirt Shadow Stripe					24.99	1			
(2384) Fishtail Flute Skirt with Free Skinny Belt Beige							24.99	1	
(2384) Fishtail Peplum Hem Pencil Skirt Black			24.99	1					
(6239) Waist Detail Pencil Skirt Suit Shadow Stripe Black							55.99	1	
(6240) Dip Hemmed Flute Skirt Suit Pinstripe Black							127.98	2	1
(6242) Function Cocktail Lace Bolero Skirt Suit Black					95.99	1	191.98	2	1
(6249-2) Smart Satin Evening Suit with Flute Skirt Beige					151.98	2	75.99	1	
(6249-3) Teddy Style Function Suit with Pencil Skirt Grey					75.99	1	151.98	2	1
(6251) Smart Casual Tailored Trouser Suit Dark Grey									1
(6252) Pinstripe Tailored Boot Leg Trouser Suit			57.99	1	57.99	1	57.99	1	1
(CH133NL) A Medium Full Bridal Petticoat Crinoline Slip Skirt Sz Large					89.97	3	89.97	3	1
(CH133NM) A Medium Full Bridal Petticoat Crinoline Slip Skirt Sz Medium					29.90	1	59.80	2	
(ONE) 1 Satin Headband			13.98	2	6.99	1	6.99	1	
(One) CHEER Rhinestone Studded Stretch Headband					5.99	1	5.99	1	
(Set of 3) Leopard Animal Print Stretch Headband							18.00	2	
(Set of 4) 2.5 Glitter Cotton Stretch Headbands					10.99	1	21.98	2	
?Ã?Â¿Long Sleeve Striped V-Neck Hoodie Pullover							97.00	6	1
[BLANKNYC] Women's Happy Trail Denim Shorts			115.83	3					1
Total	**222,578.76**	**3670**	**750,004.33**	**12715**	**1,450,147.88**	**24314**	**2,485,651.84**	**42308**	**5,120,1**

Filters

Page 1 | Page 2 | Page 3 | Page 4 | Page 5 | Page 6 | +



23.48K
Total Sales

26
Order Count

12.31K
Sum of Profit

category	Order Count	Total Sales
Active	10	9,030.00
Fashion Hoodies & Sweatshirts	9	8,127.00
Outerwear & Coats	7	6,321.00
Total	**26**	**23,478.00**

Total Sales by category

Page 1 | Page 2 | Page 3 | Page 4 | Page 5 | Page 6 | +